Roy Ho

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EDUCATION

UNIVERSITY OF CALIFORNIA, DAVIS
Bachelor of Science **September 2022 – March 2026**
Major in Statistical Data Science and Minor in Computer Science

Relevant Coursework: Statistical Learning, Regression, Probability Theory, Advanced Computing, Algorithm Design & Analysis, Data Structures

Certifications: NVIDIA – Fundamentals of Deep Learning **October 2025**

EXPERIENCE

Journal of Artificial Intelligence and Knowledge Engineering (JAIKE)
Artificial Intelligence Researcher **January 2025 – Present**
- Conducted research on LLM-based automation covering API-driven system design, agent orchestration frameworks, retrieval-augmented generation (RAG), and a consolidated threat model spanning prompt injection, data leakage, and unintended action execution.
- First-authored a 29-page, 36-source review introducing a two-axis framework for LLM agent autonomy and oversight, plus a cost model pricing oversight against productivity gains; revise-and-resubmit with publication recommended.
- Served as a peer reviewer for JAIKE, evaluating submissions on retrieval methods in large language models, reasoning performance in extended tasks, and large-scale model architectures for methodological rigor and evaluation quality.

TechSprint Innovators
Head of Data Engineering **March 2024 – September 2025**
- Built a multi-factor stock screening model using fundamental, technical, and NLP-based sentiment features (FinBERT).
- Developed a scikit-learn classification model predicting price appreciation, with feature selection, tuning, and validation on historical market data.
- Engineered and automated a daily ETL (Extract, Transform, Load) data pipeline (Python, yfinance, Alpaca API) running on a Raspberry Pi to filter equities, generate structured CSV outputs, and deliver real-time investment signals via Discord webhook.

AI Student Collective (AISC)
General Member **September 2024 – April 2025**
- Delivered machine learning and computer vision projects including stroke risk prediction and real-time drowsy driver detection.
- Performed data preprocessing, feature engineering, model development, evaluation, and project presentations.
- Worked within quarter-long sprint cycles with defined milestones, code reviews, and final project demos, from ideation to deployment.

SIGNIFICANT PROJECTS

San Francisco Restaurant Safety Map **April 2026 – May 2026**
- Built a full-stack web app mapping 20,000+ health inspections across 7,700+ San Francisco restaurants using the DataSF public feed.
- Engineered a Python Extract, Transform, Load (ETL) pipeline (pandas, geopy) normalizing inspection records into a 3-table SQLite schema with cached geocoding, then containerized the API and frontend with Docker Compose using a multi-stage Node-to-nginx build.
- Developed a Flask REST API with 7 endpoints and a React + Mapbox frontend featuring debounced search, neighborhood typeahead, geolocation-based "Near Me" centering, and an Insights panel surfacing per-neighborhood best/worst rankings via SQL window functions.

Job Market Analytics Dashboard **September 2025 – December 2025**
- Owned the data acquisition layer for a group capstone dashboard, writing JobSpy and Selenium scrapers that collected 23,000+ job posting records across seven role families in CA, NY, and TX, deduplicated to 6,800+ unique postings from 2,300+ companies.
- Built the resume-to-job matching engine as a weighted ranking model combining Term Frequency-Inverse Document Frequency (TF-IDF) cosine similarity, skill overlap against a 366-term taxonomy, and an experience-proximity bonus.
- Implemented eligibility pre-filters that stripped senior-level titles and regex-extracted required years of experience from descriptions to drop postings above the candidate's level, then delivered the ranked output into the final dashboard.

Drowsy Driver Detection System **January 2025 – March 2025**
- Owned the eye-state detection layer of a group real-time drowsy driver detection project, awarded Best Execution for the AISC Winter 2025 cycle.
- Computed Eye Aspect Ratio from dlib's 68-point facial landmark model and frontal face detector via scipy Euclidean distance, flagging drowsiness when EAR held below a 0.25 threshold across 20 consecutive frames.
- Integrated the detector into an OpenCV pipeline with grayscale preprocessing and convex hull eye overlays, adding a pygame audio alert and graceful failure handling that surfaces setup instructions when model assets are missing.

SKILLS

Programming: Python, R, SQL, JavaScript, HTML, CSS
Data Analysis & Machine Learning: Excel (advanced formulas, pivot tables, VLOOKUP, data visualization), Regression (Lasso, Ridge, Logistic), Random Forest, Gradient Boosting, CNNs, Feature Engineering, Cross-Validation & Hyperparameter Tuning, Model Evaluation (ROC-AUC, Precision/Recall), TF-IDF, NLP (spaCy)
Libraries & Frameworks: scikit-learn, pandas, NumPy, PyTorch, Flask, SQLAlchemy, SQLite, React, Vite, Mapbox GL JS, BeautifulSoup, Selenium, JobSpy, PySpark, Matplotlib, geopy, scipy, dlib, OpenCV
Tools & Platforms: Tableau, Power BI, Docker, Git, GitHub, Jupyter Notebook, VS Code, Cursor, Raspberry Pi